

5 Aerobic respiration is a metabolic process involving a series of chemical reactions.

(a) How is water formed during aerobic respiration?

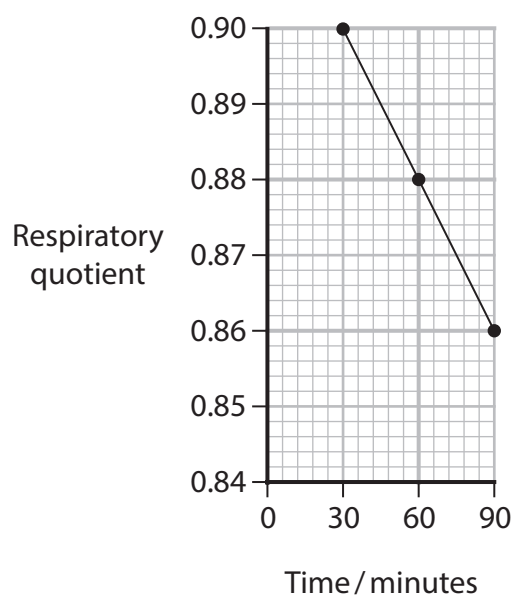
(1)

- ☐ **A** reduction of carbon
- ☐ **B** reduction of oxygen
- ☐ **C** phosphorylation of ADP
- ☐ **D** phosphorylation of glucose

(b) The effect of exercise on the respiratory quotient of a cyclist was investigated.

The cyclist undertook moderate exercise for 90 minutes.

The graph shows the respiratory quotient for the cyclist during this investigation.



(i) The rate of oxygen uptake by the cyclist at 60 minutes was $2.20 \text{ dm}^3 \text{ min}^{-1}$.

Which is the rate of carbon dioxide production by the cyclist at 60 minutes?

(1)

☐ **A** $1.30 \text{ dm}^3 \text{ min}^{-1}$

☐ **B** $1.94 \text{ dm}^3 \text{ min}^{-1}$

☐ **C** $2.50 \text{ dm}^3 \text{ min}^{-1}$

☐ **D** $3.08 \text{ dm}^3 \text{ min}^{-1}$

(ii) The table gives the respiratory quotient for aerobic respiration using fats or carbohydrates as respiratory substrates.

Substrate	Respiratory quotient
Fat	0.7
Carbohydrate	1.0

Explain the changes in respiratory quotient in this person during the exercise period of this investigation.

Use the information in the table and the graph to support your answer.

(2)

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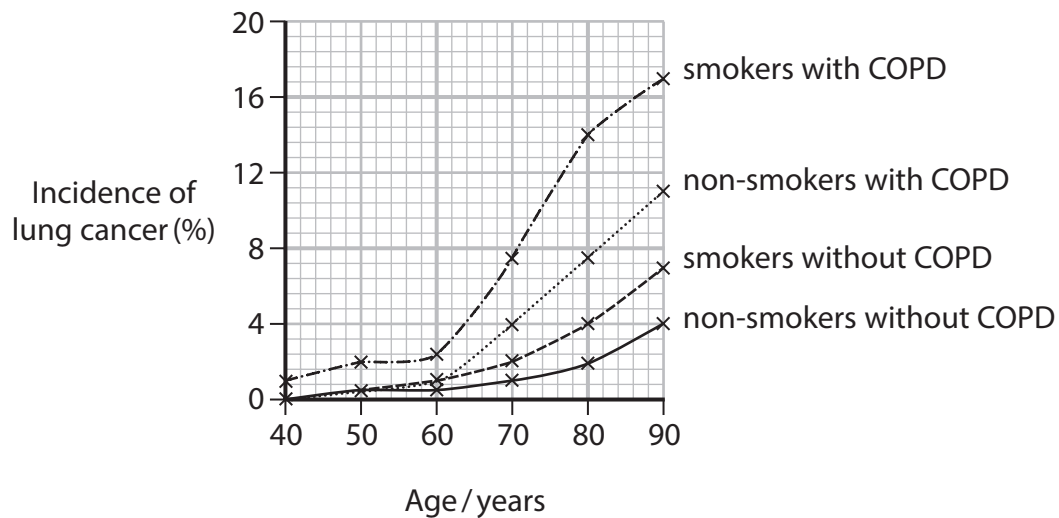
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(c) Chronic obstructive pulmonary disease (COPD) is a life-threatening illness.

This disease greatly reduces the ability to take in oxygen in the lungs and is a risk factor for lung cancer.

The graph compares the incidence of lung cancer in smoking and non-smoking individuals with and without COPD.



In South Korea there are 105 500 males and 158 600 females in the 80-year-old age group.

- (i) Calculate the total number of people in this age group that will have lung cancer who are non-smokers with COPD.

Give your answer in standard form.

(2)

Answer

- (ii) Cigarette smoking causes inflammation in the lungs. This can result in damage to the walls of the alveoli and lead to COPD.

Cigarette smoking also causes the production of thick mucus in the lungs.

Deduce the effects on the heart rate and ventilation rate in a person who smokes and has COPD.

(4)

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(Total for Question 5 = 10 marks)